

91. The complex II of mitochondrial electron transport chain is also known as
- (1) Cytochrome *bc₁*
 - (2) Succinate dehydrogenase
 - (3) Cytochrome *c* oxidase
 - (4) NADH dehydrogenase
92. Polymerase chain reaction (PCR) amplifies DNA following the equation.
- (1) N^2
 - (2) 2^n
 - (3) $2n + 1$
 - (4) $2N^2$
93. What are the potential drawbacks in adoption of the IVF method?
- A. High fatality risk to mother
 - B. Expensive instruments and reagents
 - C. Husband/wife necessary for being donors
 - D. Less adoption of orphans
 - E. Not available in India
 - F. Possibility that the early embryo does not survive
- Choose the correct answer from the options given below:
- (1) B, D, F only.
 - (2) A, C, D, F only
 - (3) A, B, C, D only
 - (4) A, B, C, E, F only
94. What is the name of the blood vessel that carries deoxygenated blood from the body to the heart in a frog?
- (1) Aorta
 - (2) Pulmonary artery
 - (3) Pulmonary vein
 - (4) Vena cava
95. Which one of the following statements refers to Reductionist Biology?
- (1) Physico-chemical approach to study and understand living organisms.
 - (2) Physiological approach to study and understand, living organisms.
 - (3) Chemical approach to study and understand living organisms.
 - (4) Behavioural approach to study and understand living organisms.

96. Given below are two statements:
- Statement I:** In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.
- Statement II:** DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.
- In the light of the above statements, choose the **most appropriate** answer from the options given below:
- (1) Both statement I and statement II are correct
 - (2) Both statement I and statement II are incorrect
 - (3) Statement I is correct but statement II is incorrect
 - (4) Statement I is incorrect but statement II is correct
97. Epiphytes that are growing on a mango branch is an example of which of the following?
- (1) Commensalism
 - (2) Mutualism
 - (3) Predation
 - (4) Amensalism
98. From the statements given below choose the **correct** option:
- A. The eukaryotic ribosomes are 80S and prokaryotic ribosomes are 70S.
 - B. Each ribosome has two sub-units.
 - C. The two sub-units of 80S ribosome are 60S and 40S while that of 70S are 50S and 30S.
 - D. The two sub-units of 80S ribosome are 60S and 20S and that of 70S are 50S and 20S.
 - E. The two sub-units of 80S are 60S and 30S and that of 70S are 50S and 30S
- (1) A, B, C are true
 - (2) A, B, D are true
 - (3) A, B, E are true
 - (4) B, D, E are true
99. Which one of the following is an example of ex-situ conservation?
- (1) National Park
 - (2) Wildlife Sanctuary
 - (3) Zoos and botanical gardens
 - (4) Protected areas.

100. Given below are two statements:

Statement I: The primary source of energy in an ecosystem is solar energy.

Statement II: The rate of production of organic matter during photosynthesis in an ecosystem is called net primary productivity (NPP).

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

101. Match List-I with List-II.

List-I		List-II	
A.	Emphysema	I.	Rapid spasms in muscle due to low Ca^{++} in body fluid
B.	Angina Pectoris	II.	Damaged alveolar walls and decreased respiratory surface
C.	Glomerulo-nephritis	III.	Acute chest pain when not enough oxygen is reaching to heart muscle
D.	Tetany	IV.	Inflammation of glomeruli of kidney

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-III, B-I, C-II, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-II, B-III, C-IV, D-I

102. Given below are two statement: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Both wind and water pollinated flowers are not very colourful and do not produce nectar.

Reason (R): The flowers produce enormous amount of pollen grains in wind and water pollinated flowers.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
- (2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.
- (3) **A** is true but **R** is false.
- (4) **A** is false but **R** is true.

103. Which of the following is an example of non—distilled alcoholic beverage produced by yeast?

- (1) Whisky
- (2) Brandy
- (3) Beer
- (4) Rum

104. Given below are two statements:

Statement I: In a floral formula $\oplus$ stands for zygomorphic nature of the flower, and $\underline{G}$ stands for inferior ovary.

Statement II: In a floral formula $\oplus$ stands for actinomorphic nature of the flower and $\underline{G}$ stands for superior ovary.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

105. Streptokinase produced by bacterium *Streptococcus* is used for

- (1) Curd production
- (2) Ethanol production
- (3) Liver disease treatment
- (4) Removing clots from blood vessels

- 106.** Which chromosome in the human genome has the highest number of genes?
- Chromosome X
 - Chromosome Y
 - Chromosome 1
 - Chromosome 10
- 107.** Which of the following statement is correct about location of the male frog copulatory pad?
- First and Second digit of fore limb
 - First digit of hind limb
 - Second digit of fore limb
 - First digit of the fore limb
- 108.** Which one of the following phytohormones promotes nutrient mobilization which helps in the delay of leaf senescence in plants?
- Ethylene
 - Abscissic acid
 - Gibberellin
 - Cytokinin
- 109.** While trying to find out the characteristic of a newly found animal, a researcher did the histology of adult animal and observed a cavity with presence of mesodermal tissue towards the body wall but no mesodermal tissue was observed towards the alimentary canal. What could be the possible coelome of that animal?
- Acoelomate
 - Pseudocoelomate
 - Schizocoelomate
 - Spongocoelomate

110. Match List-I with List-II.

List-I	List-II
A. Head	I. Enzymes
B. Middle piece	II. Sperm motility
C. Acrosome	III. Energy
D. Tail	IV. Genetic material

Choose the **correct** answer from the options given below:

- A-IV, B-III, C-I, D-II
- A-IV, B-III, C-II, D-I
- A-III, B-IV, C-II, D-I
- A-III, B-II, C-I, D-IV

111. Given below are the stages in the life cycle of pteridophytes. Arrange the following stages in the correct sequence.

- Prothallus stage
- Meiosis in spore mother cells
- Fertilisation
- Formation of archegonia and antheridia in gametophyte.
- Transfer of antherozoids to the archegonia in presence of water.

Choose the **correct** answer from the options given below:

- B, A, D, E, C
- B, A, E, C, D
- D, E, C, A, B
- E, D, C, B, A

112. Cardiac activities of the heart are regulated by:

- Nodal tissue
- A special neural centre in the medulla oblongata
- Adrenal medullary hormones
- Adrenal cortical hormones

Choose the **correct** answer from the options given below:

- A, B and C Only.
- A, B, C and D
- A, C and D Only
- A, B and D Only

113. Which of following organisms *cannot* fix nitrogen?

- Azotobacter*
- Oscillatoria*
- Anabaena*
- Volvox*
- Nostoc*

Choose the **correct** answer from the options given below:

- A only
- D only
- B only
- E only

114. Given below are two statements:

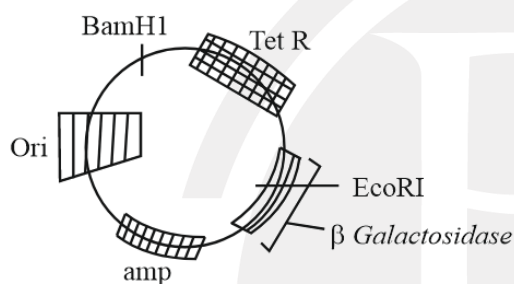
Statement I: Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II: RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

115.



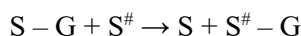
In the above represented plasmid an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

- (1) Using ampicillin & tetracyclin containing medium plate.
- (2) Blue color colonies will be selected.
- (3) White color colonies will be selected.
- (4) Blue color colonies grown on ampicillin plates can be selected.

116. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

- (1) Bacterium
- (2) Yeast
- (3) Virus
- (4) Phage

117. Name the class of enzyme that usually catalyze the following reaction:



Where, G $\rightarrow$ a group other than hydrogen

S $\rightarrow$ a substrate

S[#] $\rightarrow$ another substrate

- (1) Hydrolase
- (2) Lyase
- (3) Transferase
- (4) Ligase

118. Find the statement that is **NOT** correct with regard to the structure of monocot stem.

- (1) Hypodermis is parenchymatous.
- (2) Vascular bundles are scattered.
- (3) Vascular bundles are conjoint and closed.
- (4) Phloem parenchyma is absent.

119. The correct sequence of events in the life cycle of bryophytes is

- A. Fusion of antherozoid with egg.
- B. Attachment of gametophyte to substratum.
- C. Reduction division to produce haploid spores.
- D. Formation of sporophyte.
- E. Release of antherozoids into water.

Choose the **correct** answer from the option given below:

- (1) D, E, A, C, B
- (2) B, E, A, C, D
- (3) B, E, A, D, C
- (4) D, E, A, B, C

120. Which are correct:

- A. Computed tomography and magnetic resonance imaging detect cancers of internal organs.
- B. Chemotherapeutics drugs are used to kill non-cancerous cells.
- C. α -interferon activate the cancer patients' immune system and helps in destroying the tumour.
- D. Chemotherapeutic drugs are biological response modifiers.
- E. In the case of leukaemia blood cells counts are decreased.

Choose the **correct** answer from the option given below:

- (1) B and D only
- (2) D and E only
- (3) C and D only
- (4) A and C only

121. Match List-I with List-II.

List-I	List-II
A. Centromere	I. Mitochondrion
B. Cilium	II. Cell division
C. Cristae	III. Cell movement
D. Cell membrane	IV. Phospholipid Bilayer

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-II, C-III, D-I
- (4) A-II, B-III, C-I, D-IV

122. Match List-I with List-II.

List-I	List-II
A. Chlorophyll a	I. Yellow-green
B. Chlorophyll b	II. Yellow
C. Xanthophylls	III. Blue-green
D. Carotenoids	IV. Yellow to Yellow-orange

Choose the option with all **correct** matches.

- (1) A-III, B-IV, C-II, D-I
 (2) A-III, B-I, C-II, D-IV
 (3) A-I, B-II, C-IV, D-III
 (4) A-I, B-IV, C-III, D-II

123. Find the correct statements:

- A. In human pregnancy, the major organ systems are formed at the end of 12 weeks.
 B. In human pregnancy the major organ systems are formed at the end of 8 weeks.
 C. In human pregnancy heart is formed after one month of gestation.
 D. In human pregnancy, limbs and digits develop by the end of second month.
 E. In human pregnancy the appearance of hair usually observed in the fifth month.

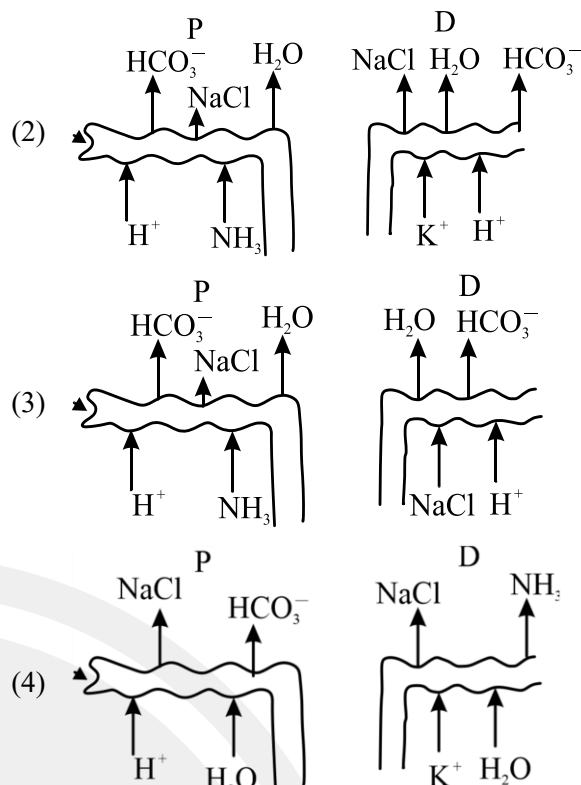
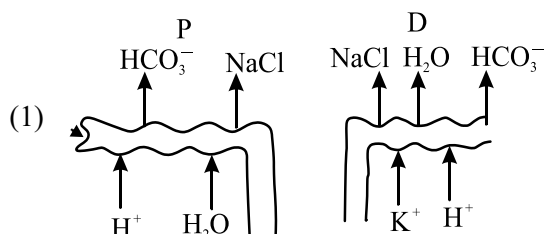
Choose the **correct** answer from the options given below:

- (1) A and E Only
 (2) B and C Only
 (3) B, C, D and E Only
 (4) A, C, D and E Only

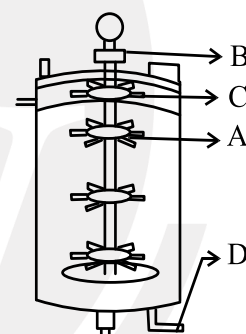
124. In the seeds of cereals, the outer covering of endosperm separates the embryo by a protein-rich layer called:

- (1) Coleoptile
 (2) Coleorhiza
 (3) Integument
 (4) Aleurone layer

125. Which of the following diagrams is correct with regard to the proximal (P) and distal (D) tubule of the Nephron.



126. Identify the part of a bio-reactor which is used as a foam braker from the given figure.



- (1) A (2) B
 (3) D (4) C

127. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): A typical unfertilized, angiosperm embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R): The egg apparatus has 2 polar nuclei. In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**.
 (2) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**.
 (3) **A** is true but **R** is false.
 (4) **A** is false but **R** is true.

128. A specialized membranous structure in a prokaryotic cell which helps in cell wall formation, DNA replication and respiration is:

- (1) Mesosome (2) Chromatophores
(3) Cristae (4) Endoplasmic Reticulum

129. Which of the following are the post-transcriptional events in an eukaryotic cell?

- A. Transport of pre-mRNA to cytoplasm prior to splicing.
B. Removal of introns and joining of exons.
C. Addition of methyl group at 5' end of hnRNA.
D. Addition of adenine residues at 3' end of hnRNA.
E. Base pairing of two complementary RNAs.

Choose the **correct** answer from the options given below:

- (1) A, B, C only (2) B, C, D only
(3) B, C, E only (4) C, D, E only

130. What is the pattern of inheritance for polygenic trait?

- (1) Mendelian inheritance pattern
(2) Non-mendelian inheritance pattern
(3) Autosomal dominant pattern
(4) X-linked recessive inheritance pattern

131. Which one of the following enzymes contains 'Haem' as the prosthetic group?

- (1) RuBisCo
(2) Carbonic anhydrase
(3) Succinate dehydrogenase
(4) Catalase

132. Each of the following characteristics represent a Kingdom proposed by Whittaker. Arrange the following in increasing order of complexity of body organization.

- A. Multicellular heterotrophs with cell wall made of chitin.
B. Heterotrophs with tissue/organ/organ system level of body organization.
C. Prokaryotes with cell wall made of polysaccharides and amino acids.
D. Eukaryotic autotrophs with tissue/organ level of body organization.
E. Eukaryotes with cellular body organization.

Choose the **correct** answer from the options given below:

- (1) A, C, E, B, D
(2) C, E, A, D, B
(3) A, C, E, D, B
(4) C, E, A, B, D

133. Who is known as the father of Ecology in India?

- (1) S. R. Kashyap
(2) Ramdeo Misra
(3) Ram Udar
(4) Birbal Sahni

134. Match List-I with List-II.

List-I		List-II	
A.	Alfred Hershey and Martha Chase	I.	Streptococcus Pneumoniae
B.	Euchromatin	II.	Densely packed and dark-stained
C.	Frederick Griffith	III.	Loosely packed and light-stained
D.	Heterochromatin	IV.	DNA as genetic material confirmation

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-IV, B-III, C-I, D-II
(4) A-III, B-II, C-IV, D-I

135. Neoplastic characteristics of cells refers to:

- A. A mass of proliferating cell
B. Rapid growth of cells
C. Invasion and damage to the surrounding tissue
D. Those confined to original location

Choose the **correct** answer from the options given below:

- (1) A, B only
(2) A, B, C only
(3) A, B, D only
(4) B, C, D only